

# Literature Status: Enflo Type and Rademacher Type

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## Status

This packet records a literature answer to open-problem signals in Giladi–Naor [1] and to the same Enflo-problem occurrence in Mendel–Naor [2]. It is not a new proof packet.

## Source Question

In the introduction of [1], on page 2 of the PDF, the authors recall Enflo’s question for Banach spaces: whether Rademacher type  $p$  implies Enflo type  $p$ . They note that Enflo type  $p$  immediately implies Rademacher type  $p$ , and that Pisier had proved the reverse implication only with exponent  $p' < p$ . Thus the endpoint implication Rademacher type  $p \Rightarrow$  Enflo type  $p$  is the open problem cited by the source.

The same classical question is recalled in [2]. That paper also asks whether scaled Enflo type  $p$  implies Enflo type  $p$ ; this packet covers only the Banach-space Enflo/Rademacher endpoint consequence, not the separate general metric-space implication.

## Supporting Answer

Ivanisvili, van Handel, and Volberg [3] explicitly settle this problem. Their abstract states that Rademacher type and Enflo type coincide, and Theorem 1.1 gives the quantitative comparison

$$T_p^R(X) \leq T_p^E(X) \leq \frac{\pi}{\sqrt{2}} T_p^R(X)$$

for every  $p \in [1, 2]$  and every Banach space  $X$ . Since the first inequality is the immediate Enflo-implies-Rademacher direction and the second is the previously open Rademacher-implies-Enflo direction, this theorem gives a full affirmative answer to the question cited in [1] and to the same Enflo-problem occurrence in [2].

## Scope

This is a complete later-literature answer to the Enflo/Rademacher type equivalence question cited in arXiv:1004.4221 and arXiv:0506215. It does not represent a new result from this run. It also does not address separate quantitative questions about the smallest scale  $m$  in the scaled Enflo type inequality, nor the general metric implication from scaled Enflo type to Enflo type.

## Search Evidence

The local cheap indexes contained no direct previous packet for arXiv:1004.4221. Web searches for phrases such as *Rademacher type and Enflo type coincide* and *Rademacher type  $p$  implies Enflo type  $p$*  returned arXiv:2003.06345 as the direct later answer. The supporting paper explicitly frames its theorem as settling Enflo’s long-standing problem.

## References

- [1] O. Giladi and A. Naor, *Improved bounds in the scaled Enflo type inequality for Banach spaces*, arXiv:1004.4221.
- [2] M. Mendel and A. Naor, *Scaled Enflo type is equivalent to Rademacher type*, arXiv:0506215.
- [3] P. Ivanisvili, R. van Handel, and A. Volberg, *Rademacher type and Enflo type coincide*, arXiv:2003.06345.